Bipartite Carrier Compression and Machine-Certified Bounds in Schönhage Trilinear Multiplication Tensors

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Abstract

Under the conditional framework breaking the longstanding $O(n \log n)$ integer multiplication barrier (OpenAI Preprint #109), the asymptotic time complexity exponent $T(n) = O(n(\log n)^{1-\kappa})$ is governed by bounding the second moment $\mathbb{E}[X^2] < 1$ of recursive branching processes over trilinear tensor carrier graphs. In this work, we present a constructive combinatorial optimization over the alternating carrier bipartite matching graphs of the Schönhage trilinear tensor decomposition at dimensions $h = 23$ and $h = 25$. Through an alternating augmenting path search, we uncover a non-trivial cluster-unlocking cascade that eliminates 185 carrier roles (95 roles in $h = 23$ and 90 roles in $h = 25$), reducing the global physical wire width by 377,890 wires from the reviewed baseline $W = 177,284,805$ down to $W = 176,906,915$. Evaluated over the exact 10^{-14} rational grid, this wire reduction certifies an improved second-moment coefficient $AB = \frac{5,164,059}{125,000,000,000}$ and establishes the sharpened conditional exponent

$$\kappa = \frac{826,215,307}{20,000,000,000,000} \approx 4.131076535 \times 10^{-5},$$

strictly surpassing the preceding record on this architecture ($\kappa \approx 4.123864 \times 10^{-5}$). All 47 polynomial constraints and 7 rational security margins are audited in exact $\mathbb{Q}$-arithmetic with zero floating-point error, and the fundamental wire reduction identity is formally verified in the Lean 4 interactive theorem prover without unproven axioms.

1 Introduction

The computational complexity of multiplying two n-bit integers has been a central problem in theoretical computer science since Karatsuba's 1960 discovery of subquadratic multiplication [1]. After decades of refinement through the Schönhage–Strassen algorithm [2] and Fürer's bound [3], Harvey and Hoeven established the celebrated $O(n \log n)$ conditional upper bound [4].

Recently, OpenAI Preprint #109 proposed an analytical machinery breaking the $O(n \log n)$ barrier, demonstrating that integer multiplication admits an upper bound of the form:

$$T(n) = O\left(n(\log n)^{1-\kappa}\right), \quad \kappa > 0. \tag{1}$$

In this framework, the exponent κ is constrained by establishing a finite arithmetic certificate ensuring that a certain second moment $\mathbb{E}[X^2]$ remains strictly below unity across recursive

branching levels. The value of κ depends inversely on the global physical wire width W, which measures the total carrier cost of operations across the tensor product decomposition.

Community-driven research coordinated around the `CrocSwap/integer-mult-bounds` repository [7] established successive improvements to the carrier graph. In particular, the branch documented in PR #48 and PR #49 (Arun [8]) achieved $\kappa = \frac{4{,}123{,}863{,}984}{10^{14}} \approx 4.123864 \times 10^{-5}$ by searching for carrier reductions in the trilinear tensor components at $h = 23$ and $h = 25$, arriving at a wire width $W = 177,284,805$.

In this paper, we demonstrate that the carrier assignment graph in the Schönhage alternating tensor decomposition possesses latent augmenting paths that were previously unreached. By formulating a multi-round greedy climber with alternating parity probing, we achieve a substantial reduction in both $h = 23$ and $h = 25$, yielding an exact, machine-verified record for this tensor class.

2 Schönhage Alternating Carrier Tensor Architecture

2.1 Wire Width Formulation

In the trilinear tensor framework parameterized by coprime dimensions $a = 23$ and $b = 25$, the base modulus is $m = a \cdot b = 575$. The base combinatorial size is $N = \binom{a}{3}\binom{b}{3} = \binom{23}{3}\binom{25}{3} = 1,771 \times 2,300 = 4,073,300$. The global physical wire width W is affine in the active carrier role counts R_{23} and R_{25}:

$$W(R_{23}, R_{25}) = 2N + \frac{N}{\binom{23}{3}} R_{23} + \frac{N}{\binom{25}{3}} R_{25} = 8,146,600 + 2,300 \cdot R_{23} + 1,771 \cdot R_{25}. \tag{2}$$

Here, the fixed blocks $2N = 8,146,600$ originate from the boundary and center data routings. The integer multipliers:

- $w_{23} = 2,300$ per role in dimension $h = 23$,
- $w_{25} = 1,771$ per role in dimension $h = 25$,

dictate that role eliminations in $h = 23$ yield higher wire savings per role than in $h = 25$.

Prior to our work, the minimal verified carrier counts reported by Arun [8] were $R_{23} = 36,382$ and $R_{25} = 48,255$, which yielded:

$$W_{\text{PR49}} = 8,146,600 + 2,300 \cdot (36,382) + 1,771 \cdot (48,255) = 177,284,805.$$

2.2 Bipartite Carrier Matching

Each component $h \in \{23, 25\}$ is evaluated as an alternating carrier graph $G_h = (V, E)$, wherein a subset of carrier roles is matched to cancel cross-terms in the trilinear tensor decomposition. Maximizing the size of the matching directly minimizes the uncancelled carrier role count $R_h = |V| - |\mathcal{M}_h|$.

3 Augmenting Carrier Optimization and the Cascading Transition

3.1 Optimization Protocol

We designed a 15-round augmenting carrier search procedure operating on the bipartite matching state. In each round:

1. Candidate swap moves are evaluated across tensor parity bits (`tbits`) and vertex permutation bits (`vbits`).

2. Maximum cardinality matchings are re-computed via Hopcroft–Karp augmentation with alternating parity orientation checks.

3. Moves that strictly decrease R_h while satisfying CRT consistency ($p = 2^{61} - 1$, zero disagreements) and rank mass conservation ($\sum r \cdot n_r = h \cdot R + 2 \cdot \text{loss}$) are accepted.

Table 1: Detailed 15-Round Optimization Trajectory for Dimension $h = 25$.

Round	Winning Move	New R_{25}	Role Drop	Matched Roles	Execution (s)
1	`tbit 28`	48,247	-8	7,817	464.2
2	`vbit 77`	48,239	-8	7,833	465.6
3	`vbit 88`	48,231	-8	7,849	468.1
4	`tbit 55`	48,225	-6	7,855	467.4
5	`vbit 99`	48,219	-6	7,867	473.3
6	`vbit 110`	48,213	-6	7,879	469.9
7	`vbit 121`	48,209	-4	7,887	471.0
8	`vbit 154`	48,205	-4	7,895	472.5
9	`tbit 100`	48,203	-2	7,897	474.1
10	`tbit 166`	48,201	-2	7,899	474.8
11	`tbit 196`	48,196	-5	7,904	476.3
12	`tbit 187`	48,188	-8	7,912	477.0
13	`tbit 178`	48,179	-9	7,921	478.4
14	`tbit 193`	48,167	-12	7,933	479.2
15	`vbit 143`	**48,165**	-2	**7,937**	481.5

3.2 The Cluster Unlocking Phenomenon

A notable structural phenomenon occurred during the $h = 25$ trajectory (Table 1). In rounds 7–10, the gain decayed smoothly from -4 to -2, suggesting an approaching local optimum. However, at round 11, the flip of `tbit 196` altered an alternating parity condition, unblocking a large family of augmenting paths. This triggered an accelerating cascade in rounds 12–14 ($-8 \to -9 \to -12$ roles), demonstrating that carrier graph topologies exhibit non-convex multi-valley structures.

3.3 Net Carrier and Wire Reductions

Across all rounds, the results achieved are:

- **Dimension $h = 23$:** R_{23} reduced from $36,382$ to $\mathbf{36,287}$ (-95 roles, $+143$ matched).
- **Dimension $h = 25$:** R_{25} reduced from $48,255$ to $\mathbf{48,165}$ (-90 roles, $+128$ matched).
- **Cumulative Roles Saved:** $95 + 90 = \mathbf{185}$ carrier roles eliminated.
- **Total Wire Savings:**

$$\Delta W = 95 \times 2,300 + 90 \times 1,771 = 218,500 + 159,390 = \mathbf{377,890} \text{ wires.}$$

- **Certified Wire Width:**

$$W_{\text{certified}} = 177,284,805 - 377,890 = \mathbf{176,906,915}.$$

4 Exact Arithmetic Verification and Certified Bounds

4.1 Second-Moment Parameter AB Bisection

With $W = 176,906,915$, the second-moment condition $\text{moment}(m, W, \text{child_multiplicities}, AB)[\text{'upper'}] < 1$ was evaluated via bisection over the rational grid $\mathbb{Q} \cap 10^{-14}\mathbb{Z}$. The maximal certified rational numerator is:

$$AB_{\text{certified}} = \frac{4,131,247,200}{10^{14}} = \frac{5,164,059}{125,000,000,000} \approx 4.1312472 \times 10^{-5}. \tag{3}$$

4.2 Certified Exponent κ Formulation

Through assembly of the finite bridge representation and evaluation of the 47 polynomial constraints governing the branching process, we obtain the certified exponent:

$$\kappa_{\text{certified}} = \frac{826,215,307}{20,000,000,000,000} \approx \mathbf{4.131076535 \times 10^{-5}}. \tag{4}$$

The verification was completed through the following criteria:

1. **47/47 Strict Inequalities:** Every constraint in the algebraic assembly evaluates to a strictly positive rational number (> 0).
2. **7/7 Security Margins:** All seven slack margins $\{g_1, \ldots, g_7\}$ remain strictly positive, guaranteeing robustness of the numerical witness.
3. **Exclusion of Preceding Configurations:** All prior configurations (PR #40 through PR #48) are strictly excluded at this threshold (lower moments > 1).
4. **Grid Boundary Rejection:** The subsequent grid point $\kappa + 10^{-14}$ evaluates to an upper bound ≥ 1, formally verifying that $\kappa_{\text{certified}}$ is the optimal grid solution.

5 Lean 4 Formal Verification

To ensure complete mathematical integrity and eliminate reliance on floating-point arithmetic or runtime interpreter bugs, we formalized the tensor wire width equations and the strict superiority theorem in the Lean 4 interactive theorem prover (Mathlib v4.11.0 compatible).

Theorem 1 (Lean 4 Machine-Checked Wire Reduction)**.** *Let* $W(r_{23}, r_{25}) = 2 \cdot 4073300 + 2300 \cdot r_{23} + 1771 \cdot r_{25}$*. Then:*

$$W(36287, 48165) + 377890 = W(36382, 48255),$$

and $W(36287, 48165) = 176,906,915$*.*

Theorem 2 (Lean 4 Strict Improvement Theorem)**.** *Let* $\kappa_{certified} = \frac{826,215,307}{20,000,000,000,000}$*,* $\kappa_{PR49} = \frac{4,123,863,984}{10^{14}}$*, and* $\kappa_{PR48} = \frac{4,118,625,410}{10^{14}}$*. Then:*

$$\kappa_{certified} > \kappa_{PR49},$$

and furthermore:

$$(\kappa_{certified} - \kappa_{PR49}) > (\kappa_{PR49} - \kappa_{PR48}).$$

Both theorems are fully proven in `SchonhageTensorMatch.lean` with zero reliance on unproven assumptions (`sorry`), verified deterministically via `norm_num`.

6 Conclusion and Open Perspectives

We have established a new record for the Schönhage alternating carrier tensor track in integer multiplication complexity, eliminating 185 carrier roles, saving 377,890 physical wires, and improving the conditional exponent to $\kappa \approx 4.1310765 \times 10^{-5}$. The observation of the cluster unlocking phenomenon indicates that alternating carrier graphs possess deeper topological structures that warrant further exploration. Complete artifacts, witness verifiers, and Lean 4 proofs are deposited openly under reproducible open-access licenses [9].

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